

Artificial Intelligence is increasingly becoming a core layer in modern healthcare systems, transforming how diseases are diagnosed, treated, and managed. Machine learning models are now widely used to analyze medical imaging such as X-rays, CT scans, and MRI data, helping doctors detect conditions like tumors, fractures, and neurological disorders at an earlier stage with higher consistency. In clinical decision support systems, AI assists physicians by processing large volumes of patient records, lab results, and medical histories to suggest possible diagnoses or treatment plans, reducing cognitive load and improving accuracy. Natural language processing is also being applied to extract meaningful information from unstructured clinical notes, discharge summaries, and research papers, making healthcare data more usable. Beyond hospitals, AI-powered predictive analytics helps public health organizations forecast disease outbreaks and optimize resource allocation, especially during emergencies. However, the adoption of AI in healthcare also raises challenges such as data privacy concerns, bias in training datasets, regulatory compliance, and the need for interpretability in critical medical decisions. Despite these limitations, the integration of AI with traditional healthcare workflows is steadily improving efficiency, reducing operational costs, and enabling more personalized and preventive care approaches for patients across different regions and healthcare systems.